and spacetime, more generally—is substantial enough to provide the ultimate reference for motion. The verdict? Although the issue is still debated, as we've now seen, the most straightforward reading of Einstein and his general relativity is that spacetime can provide such a benchmark: *spacetime is a some*thing.[23]

Notice, though, that this conclusion is also cause for celebration among supporters of a more broadly defined relationist outlook. In Newton's view and subsequently that of special relativity, space and then spacetime were invoked as entities that provide the reference for defining accelerated motion. And since, according to these perspectives, space and spacetime are absolutely unchangeable, this notion of acceleration is absolute. In general relativity, though, the character of spacetime is completely different. Space and time are dynamic in general relativity: they are mutable; they respond to the presence of mass and energy; they are not absolute. Spacetime and, in particular, the way it warps and curves, is an embodiment of the gravitational field. Thus, in general relativity, acceleration relative to spacetime is a far cry from the absolute, staunchly un-relational conception invoked by previous theories. Instead, as Einstein argued eloquently a few years before he died,[24] acceleration relative to general relativity's spacetime *is* relational. It is not acceleration relative to material objects like stones or stars, but it is acceleration relative to something just as real, tangible, and changeable: a field—the gravitational field.[6] In this sense, spacetime—by being the incarnation of gravity—is *so* real in general relativity that the benchmark it provides is one that many relationists can comfortably accept.

Debate on the issues discussed in this chapter will no doubt continue as we grope to understand what space, time, and spacetime actually are. With the development of quantum mechanics, the plot only thickens. The concepts of empty space and of nothingness take on a whole new meaning when quantum uncertainty takes the stage. Indeed, since 1905, when Einstein did away with the luminiferous aether, the idea that space is filled with invisible substances has waged a vigorous comeback. As we will see in later chapters, key developments in modern physics have reinstituted various forms of an aetherlike entity, none of which set an absolute standard for motion like the original luminiferous aether, but all